

Server

Technical Datasheet Revision 4.0

Node-RED application for MQTT ingestion, integrity validation, alarm routing, topology, datasets, and commands.



TARGET HOST	RUNTIME	DATA SECURITY	ENGINEERING DATA
Customer VM	Node-RED	AES-128-GCM	MySQL / CSV

Key features

- Receives and separates uplink, status, topology, and integrity-path data.
- Validates frames before data enters the normal application path.
- Routes authenticated alarm events and separates manual/test input.
- Builds authenticated high-level mode commands.

Technical overview

The Server provides an application-side data and command pipeline targeted for deployment on a customer VM.

Functional profile

Runtime	Node-RED
Ingress	MQTT
Integrity	CRC + AES-GCM
Target	Customer VM

Operational interfaces

Transport	MQTT through the broker selected for deployment.
Integrity	Frame structure, length, CRC, authentication, semantic checks, and replay policy.
Application	Decoded data, alarm routing, topology, engineering UI, and commands.

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1 Product identification

Product	Server
Product status	Engineering Prototype
Document revision	4.0
Technical baseline	Server Application Protocol 0.2.0
Issuer	Lab IoT ITB

1.1 Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
AES-GCM	Authenticated encryption using Galois/Counter Mode	NVM	Non-volatile memory/state
CA	Certificate Authority	TLS	Transport Layer Security
CRC	Cyclic Redundancy Check	VM	Virtual machine on customer infrastructure
MQTT	Message Queuing Telemetry Transport	UI	User interface

2 Overview and architecture

2.1 Application overview

The Server provides an application-side data and command pipeline targeted for deployment on a customer VM.

Runtime	Node-RED	Ingress	MQTT
Integrity	CRC + AES-GCM	Target	Customer VM

- Receives and separates uplink, status, topology, and integrity-path data.
- Validates frames before data enters the normal application path.
- Routes authenticated alarm events and separates manual/test input.
- Builds authenticated high-level mode commands.

2.2 Server logical architecture

The pipeline separates transport, integrity, application routing, storage, and command downlink.



Figure 1. Server application logical pipeline.

Transport	MQTT through the broker selected for deployment.
Integrity	Frame structure, length, CRC, authentication, semantic checks, and replay policy.
Application	Decoded data, alarm routing, topology, engineering UI, and commands.

2.3 MQTT input categories

Inputs are functionally separated so status, topology, payload, and integrity events are not treated as the same data.

Category	Purpose	Downstream path
Device uplink	Carries frames/data from the field	Validation and decode
Gateway status	Gateway communication state	Monitoring/status
Topology	Parent, discovery, and route information	Engineering topology view
Integrity/raw	Validation failures or observability	Quarantine/integrity path

Note: Each input category is mapped to a separate processing path.

3 Frame validation and security

3.1 Frame validation

A frame must pass structural checks before authentication and semantic processing.

Stage	Description
Receive	Message received from MQTT
Structure	Format and mandatory fields
Length	Frame/payload size is checked
CRC	Transport integrity is checked
Security/semantic	Enters the authentication gate

Figure 2. Frame-validation gate sequence.

- Frames failing structure/length/CRC do not become decoded production data.
- Validation errors can still be routed to an integrity/diagnostic path.

3.2 AES-128-GCM authentication

GasleakDetector payloads are authenticated and decrypted with a required runtime key.

Stage	Description
Encrypted payload	Received after the frame gate
Runtime key	Required and mapped to key ID
AES-128-GCM	Decrypt + authentication-tag check
Semantic validation	Identity/role and value constraints
Valid output	Only then enters application routing

Figure 3. Field-payload security gate.

Note: A runtime key is required; a payload is rejected when authentication cannot be completed.

3.3 Replay, nonce reuse, and quarantine

Authenticated but repeated data can still be rejected so old events are not reprocessed as new data.

Condition	Decision	Path
Authentication failure	Reject	Integrity/quarantine
Replay detected	Reject	Integrity/quarantine
Nonce reuse	Reject	Integrity/quarantine
Semantic invalid	Reject	Integrity/quarantine
Valid and fresh	Accept	Decoded/application

Anti-replay state	The Server deployment must configure persistent storage.
Service recovery	With that configuration, anti-replay state is reloaded after a service restart.

4 Alarm, topology, and commands

4.1 Alarm and event handling

The Server routes alarms only after a frame passes integrity and authentication gates.

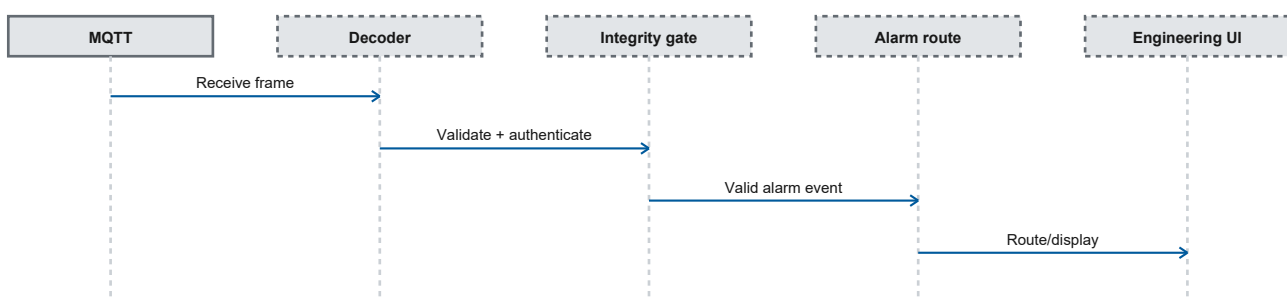


Figure 4. Server alarm routing.

Input	Production alarm path
Authenticated field alarm	May be routed
Manual/test input	Separated
Failed authentication/replay	Quarantined; not a production alarm

Note: The Server produces application alarm events; external notification integration uses customer-deployment interfaces.

4.2 Engineering topology view

The Server provides a view of Gateway, CH, GasleakDetector, active parent, discovery candidates, and routes.

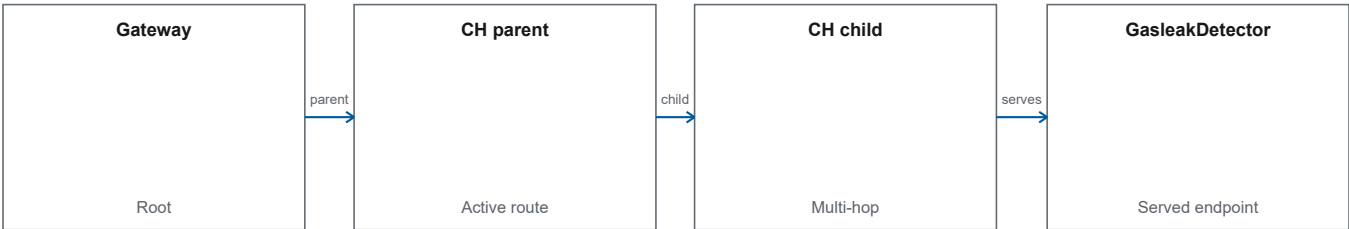


Figure 5. Information available in the topology view.

- The view can show installed parent, discovery candidates, and route lineage.
- The topology flow is part of the Node-RED-based Server application.
- Gateway and network configuration is adapted for each customer deployment.

4.3 High-level mode command

The Server can build an authenticated operational-mode command for forwarding toward GasleakDetector.

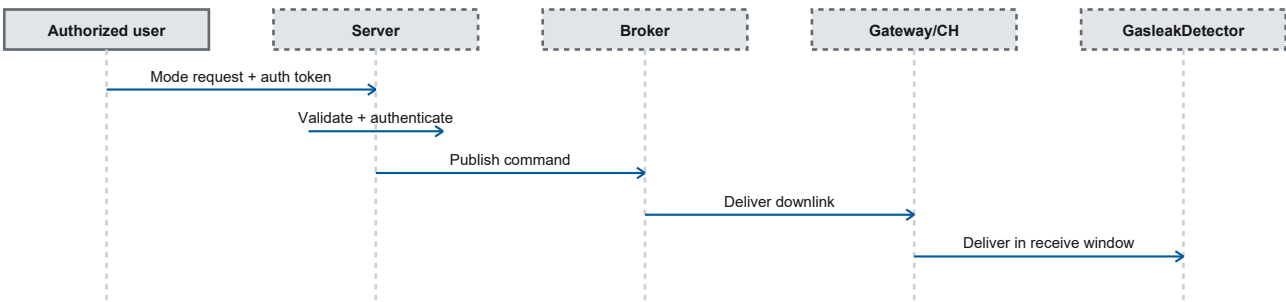


Figure 6. Server-to-field command path.

- External commands require an authorization token.
- Command payload is authenticated at the application layer.
- Current scope is high-level mode change, not general remote control.

5 Broker, storage, and deployment

5.1 Server-to-broker TLS connection

The generated flow supports a Node-RED-to-broker connection using credentials and CA-verified TLS.

Stage	Description
Node-RED	MQTT client
Credentials	Deployment-provided
CA trust	Broker certificate verification
TLS session	Server/Node-RED to broker
Broker	Trust boundary

Figure 7. Server-to-broker TLS session.

Note: This is separate from the Gateway-to-broker TLS session. It must not be described as end-to-end TLS from GasleakDetector to Server.

5.2 Dataset persistence: MySQL and CSV

The engineering dataset recorder validates records, writes to MySQL when configured, and creates a CSV copy for every accepted record.

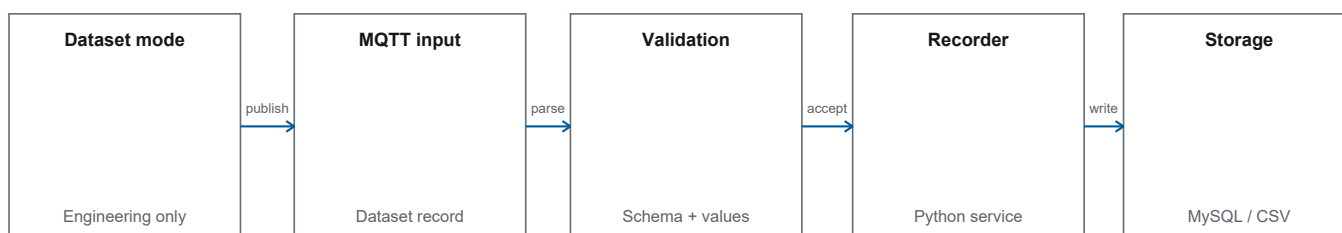


Figure 8. Engineering dataset path.

Storage	Status	Boundary
MySQL	Supported when the driver and connection are available	Dataset records only
CSV	Parallel copy; remains available when MySQL is unavailable	Engineering dataset

Note: MySQL is not generalized as storage for all telemetry, alarms, topology, or operational events.

5.3 Customer VM deployment model

The application is targeted for installation on a customer-provided virtual machine.

Area	Deployment specification
Host	VM on customer infrastructure
OS	Linux or Windows, selected for the deployment
Security	Runtime secrets and broker trust are required

Note: VM, network, broker, and secret configuration is established for each customer deployment.

5.4 Web and engineering UI

The Server provides a Node-RED-based engineering topology view and supports a web application on customer infrastructure.

UI layer	Implementation	Scope
Node-RED engineering view	Part of the Server runtime	Topology, parent, discovery, route, engineering status
Customer web application	Hosted on customer infrastructure	Web-based interface application

- Authentication, roles, browser support, and network integration are established during deployment.
- The topology view displays parent, discovery, route, and engineering status.

5.5 Deployment and operational responsibilities

Server readiness depends on the boundary between the Lab IoT ITB application and customer infrastructure.

Area	Required input/ownership
VM	Provisioning, OS image, patching, backup, resource monitoring
Broker	FQDN/IP, port, credentials, CA, ACL, retention policy
Database	MySQL endpoint/credentials/schema ownership for datasets
Secrets	Runtime AES key, command authorization token, MQTT credentials
Operations	Log collection, restart policy, incident response, and log retention

Note: VM, broker, database, secret, and log operations are managed under customer infrastructure policy.

6 Data model, storage, and security

6.1 Application data model

The Server converts device data into application fields used by storage, topology display, and alarms.

Field	Meaning	Use
node ID	Source-node identifier.	Device and topology correlation.
gas class / gas name	Classification result and its display name.	Detection status in the application.
confidence	Inference-result confidence value.	Context for result interpretation.
battery mV	Battery voltage reported by the device.	Power monitoring.
alarm	Device alarm state.	Alarm indication in the application interface.
external power	External power-source state.	Power-supply diagnostics.
sequence	Source message sequence number.	Ordering and repetition detection.
gateway	Gateway that received the data.	Communication-path correlation.
last seen	Time at which data was last received.	Connectivity state in the application.

6.2 Dataset record schema

The dataset recorder stores identity, acquisition context, eight sensor values, gain, sensor status, and an idempotency key in each record.

Group	Fields	Description
Identity	Device, node, and unique-record identity	Source identity and unique record key.
Context	Mode, sequence, time, label, and nulling profile	Acquisition context for each record.
Sensor voltages	Eight validated sensor values	One value for each channel.
Gain	Per-channel gain	Acquisition gain for each channel.
Sensor health	Eight-channel validity state	Validity state for each sensor channel.
Feature order	Canonical sensor order	Preserves dataset-structure consistency.

Note: Each record has a unique 64-character identifier so that resubmission of the same record does not create a duplicate in primary storage.

6.3 Dataset admission rules

A record is accepted only when all data-integrity requirements are satisfied before storage.

Check	Requirement
Mode	The record must use engineering dataset mode.
Source identity	The device identity must be valid, consistent with the ingress path, and match the record node identity.
Sequence and time	Sequence and timestamp must be within the unsigned 32-bit integer range.
Nulling profile	The profile identifier is within the 1-255 range.
Label	Required, limited to safe characters, and no more than 31 characters.
Sensor-value count	Exactly eight voltage values, all finite.
Gain	Each gain is one of 1, 2, 4, 8, 16, 32, or 64.
Sensor status	All sensor status values indicate valid.
Feature order	Must match the canonical MQ8, MQ135, MQ3, MQ5, MQ4, MQ7, MQ6, MQ2 order.
Idempotency	The record key is checked before writing so that the same record is not duplicated.

6.4 Storage and service resilience

Primary storage uses MySQL. Every accepted record is also written as a parallel CSV copy, which remains available when MySQL is unavailable.

Component	Role	Behavior
MySQL Server	Primary storage	Enforces unique record identifiers before data is accepted as a new record.
CSV	Parallel copy	Written for every accepted record and remains a storage path when MySQL is unavailable.
Web application	Engineering interface	Displays device data, alarms, and topology from Server services.

6.5 Deployment security requirements

The Server is installed on customer infrastructure. The following controls apply when services are not restricted to a loopback interface.

- Broker connections use TLS and authentication consistent with customer policy.
- Credentials and cryptographic material are supplied through the runtime environment or a secret store, not embedded in application files.
- The deployment must configure persistent anti-replay storage so protection can be restored after a service restart.
- Database, broker, and web-application access is restricted by customer network controls and service accounts.
- Backup, retention, operating-system patching, and VM monitoring form part of customer infrastructure operations.

Area	Server application behavior	Deployment responsibility
VM runtime	Runs the Node-RED services and application interface.	The customer provides the VM and selects Linux or Windows.
MQTT broker	Establishes an authenticated session with TLS verification.	The customer provides the broker endpoint, CA, credentials, and access policy.
Storage	Validates records, writes MySQL, and creates the parallel CSV copy.	The customer manages the database service, access, backup, and retention.
Anti-replay	Loads and updates the configured persistent replay state.	The customer provides persistent storage at deployment.
Web access	Provides the engineering interface for data, alarms, and topology.	The customer restricts network access, service accounts, and interface exposure.

7 References and revision history

Protocol and runtime references support the Server implementation specification.

Reference	Document
R1	Node-RED - User Guide, nodered.org/docs/user-guide/
R2	OASIS - MQTT Version 3.1.1, OASIS Standard
R3	Oracle - MySQL 8.4 Reference Manual
R4	NIST SP 800-38D - Recommendation for Galois/Counter Mode (GCM)

Rev	Change
3.0	Updated technical structure and specifications.